Lab Report: Kubernetes Basics with Minikube

1. Introduction

1.1 Purpose of the Lab

The purpose of this laboratory exercise is to gain hands-on experience with Kubernetes, the industry standard for container orchestration. Kubernetes automates the deployment, scaling, and management of containerized applications. It solves critical challenges such as deciding which server runs which container, ensuring containers can communicate across hosts, automatically restarting failed containers, scaling applications based on demand, and performing updates without downtime.

1.2 Learning Objectives

By completing this lab, students will be able to:

- Set up a local Kubernetes development environment using Minikube
- Create and manage Deployments to run containerized applications
- Expose applications to external traffic using Kubernetes Services
- Monitor cluster resources using both kubectl and the Kubernetes Dashboard
- Scale applications horizontally by adjusting the number of replicas
- Perform rolling updates to deploy new versions without downtime
- Enable observability features to monitor resource utilization

1.3 Required Tools

The following tools were used in this lab:

- **Minikube** (v1.35+): Creates a single-node Kubernetes cluster locally
- **kubectl** (v1.30+): Command-line interface for interacting with Kubernetes
- **Docker**: Provides container runtime capabilities for Minikube
- **Web browser**: For accessing the Kubernetes Dashboard and sample application

2. Setting Up the Kubernetes Cluster

2.1 Starting Minikube

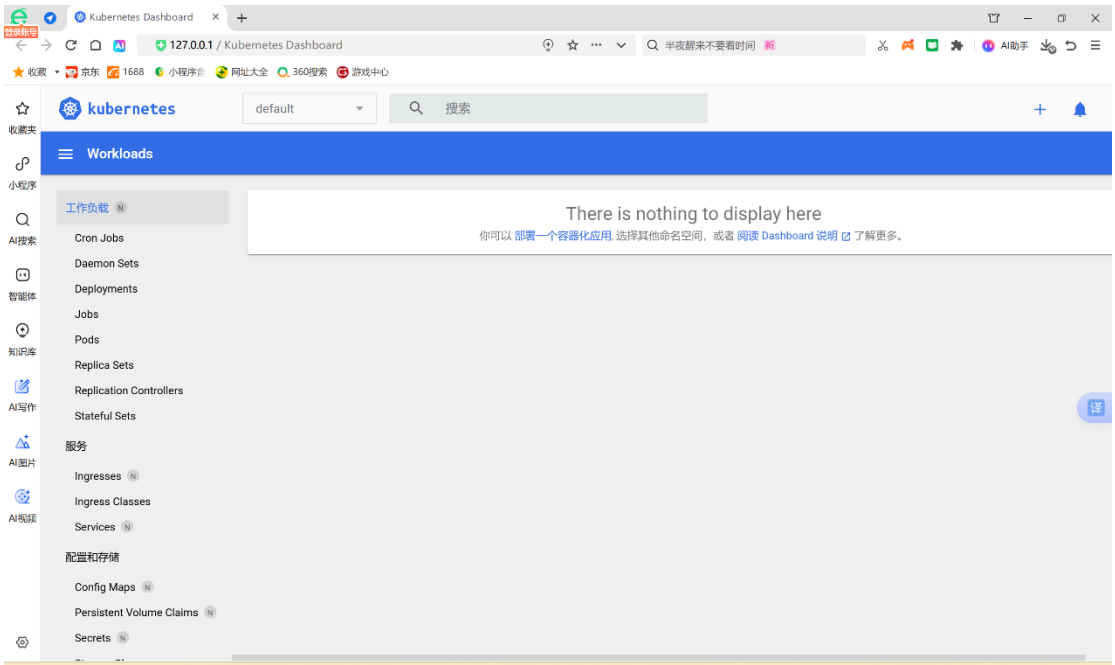
The first step was to create a functional Kubernetes cluster using Minikube with Docker as the driver. When Minikube starts, it performs several operations behind the scenes: it pulls the necessary base images containing Kubernetes components (API server, etcd, controller manager, scheduler, and kubelet), creates a Docker container to serve as the control plane node, starts all control plane components, configures networking, and updates the kubeconfig file to enable kubectl communication.

After starting the cluster, the status was verified to ensure all components were running properly. The output confirmed that the host, kubelet, and apiserver were all running, and kubectl was properly configured.

The node information was also checked. In this Minikube environment, there is a single node that serves as both the control plane and worker node. The node showed a status of "Ready", indicating it was healthy and capable of accepting workloads.

```
PS C:\Users\36237> minikube start --driver=docker
* Microsoft Windows 11 Home China 25H2 上的 minikube v1.38.1
E0419 18:07:02.134961   18524 start.go:846] api.Load failed for minikube: filestore "minikube": Docker machine "minikube
" does not exist. Use "docker-machine ls" to list machines. Use "docker-machine create" to add a new one.
E0419 18:07:02.136042   18524 start.go:846] api.Load failed for minikube: filestore "minikube": Docker machine "minikube
" does not exist. Use "docker-machine ls" to list machines. Use "docker-machine create" to add a new one.
* 根据现有的配置文件使用 docker 驱动程序
* 在集群中 "minikube" 启动节点 "minikube" primary control-plane
* 正在拉取基础镜像 v0.0.50 ...
! minikube cannot pull kicbase image from any docker registry, and is trying to download kicbase tarball from github rel
ease page via HTTP.
! 您很有可能遇到了网络问题。请确保您至少可以通过 HTTP（直连或代理）访问互联网。当前您的代理配置为：

E0419 18:09:10.069815   18524 cache.go:239] Error downloading kic artifacts:  failed to download kic base image or any f
allback image
* 创建 docker container (CPU=2, 内存=8000MB) .../
* 正在 Docker 29.2.1 中准备 Kubernetes v1.35.1…/
* 配置 bridge CNI (Container Networking Interface) ...
* 正在验证 Kubernetes 组件...
  - 正在使用镜像 gcr.io/k8s-minikube/storage-provisioner:v5
* 启用插件： storage-provisioner, default-storageclass
* 完成！ kubectl 现在已配置，默认使用"minikube"集群和"default"命名空间
```

```
PS C:\Users\36237> minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```



```
PS C:\Users\36237> kubectl create deployment hello-node --image=registry.k8s.io/e2e-test-images/agnhost:2.53 -- /agnhost netexec --http-port=8080
deployment.apps/hello-node created
PS C:\Users\36237> kubectl get deployments
NAME         READY   UP-TO-DATE   AVAILABLE   AGE
hello-node   0/1     1            0           20s
PS C:\Users\36237> kubectl get pods
NAME                          READY   STATUS    RESTARTS   AGE
hello-node-64fc5894d-9zpm8    1/1     Running   0          4m13s
PS C:\Users\36237> kubectl get events
LAST SEEN   TYPE     REASON                   OBJECT                          MESSAGE
4m25s       Normal   Scheduled                pod/hello-node-64fc5894d-9zpm8  Successfully assigned default/hello-node-64fc5894d-9zpm8 to minikub
e
4m24s       Normal   Pulling                  pod/hello-node-64fc5894d-9zpm8  Pulling image "registry.k8s.io/e2e-test-images/agnhost:2.53"
3m58s       Normal   Pulled                   pod/hello-node-64fc5894d-9zpm8  Successfully pulled image "registry.k8s.io/e2e-test-images/agnhost:
2.53" in 26.166s (26.167s including waiting). Image size: 139374622 bytes.
3m58s       Normal   Created                  pod/hello-node-64fc5894d-9zpm8  Container created
3m58s       Normal   Started                  pod/hello-node-64fc5894d-9zpm8  Container started
4m25s       Normal   SuccessfulCreate         replicaset/hello-node-64fc5894d Created pod: hello-node-64fc5894d-9zpm8
4m25s       Normal   ScalingReplicaSet        deployment/hello-node           Scaled up replica set hello-node-64fc5894d from 0 to 1
17m         Normal   Starting                 node/minikube                   Starting kubelet.
17m         Normal   NodeHasSufficientMemory  node/minikube                   Node minikube status is now: NodeHasSufficientMemory
17m         Normal   NodeHasNoDiskPressure    node/minikube                   Node minikube status is now: NodeHasNoDiskPressure
17m         Normal   NodeHasSufficientPID     node/minikube                   Node minikube status is now: NodeHasSufficientPID
17m         Normal   NodeAllocatableEnforced  node/minikube                   Updated Node Allocatable limit across pods
16m         Normal   Starting                 node/minikube                   Starting kubelet.
16m         Normal   NodeAllocatableEnforced  node/minikube                   Updated Node Allocatable limit across pods
16m         Normal   NodeHasSufficientMemory  node/minikube                   Node minikube status is now: NodeHasSufficientMemory
16m         Normal   NodeHasNoDiskPressure    node/minikube                   Node minikube status is now: NodeHasNoDiskPressure
16m         Normal   NodeHasSufficientPID     node/minikube                   Node minikube status is now: NodeHasSufficientPID
16m         Normal   NodeReady                node/minikube                   Node minikube status is now: NodeReady
16m         Normal   RegisteredNode           node/minikube                   Node minikube event: Registered Node minikube in Controller
16m         Normal   Starting                 node/minikube
```

```
PS C:\Users\36237> kubectl config view
apiVersion: v1
clusters:
- cluster:
    certificate-authority: C:\Users\36237\.minikube\ca.crt
    extensions:
    - extension:
        last-update: Sun, 19 Apr 2026 18:11:43 CST
        provider: minikube.sigs.k8s.io
        version: v1.38.1
      name: cluster_info
    server: https://127.0.0.1:53025
  name: minikube
contexts:
- context:
    cluster: minikube
    extensions:
    - extension:
        last-update: Sun, 19 Apr 2026 18:11:43 CST
        provider: minikube.sigs.k8s.io
        version: v1.38.1
      name: context_info
    namespace: default
    user: minikube
  name: minikube
current-context: minikube
kind: Config
users:
- name: minikube
  user:
    client-certificate: C:\Users\36237\.minikube\profiles\minikube\client.crt
    client-key: C:\Users\36237\.minikube\profiles\minikube\client.key
```

```
PS C:\Users\36237> kubectl logs hello-node-64fc5894d-9zpm8
I0419 10:24:40.656263       1 log.go:245] Started HTTP server on port 8080
I0419 10:24:40.656589       1 log.go:245] Started UDP server on port  8081
```

```
PS C:\Users\36237> kubectl expose deployment hello-node --type=LoadBalancer --port=8080
service/hello-node exposed
PS C:\Users\36237> kubectl get services
NAME         TYPE           CLUSTER-IP      EXTERNAL-IP   PORT(S)          AGE
hello-node   LoadBalancer   10.105.14.219   <pending>     8080:32180/TCP   19s
kubernetes   ClusterIP      10.96.0.1       <none>        443/TCP          30m
PS C:\Users\36237> minikube service hello-node
```

NAMESPACE	NAME	TARGET PORT	URL
default	hello-node	8080	http://192.168.49.2:32180

```
* 为服务 hello-node 启动隧道。
```

NAMESPACE	NAME	TARGET PORT	URL
default	hello-node		http://127.0.0.1:61494

```
* 正通过默认浏览器打开服务 default/hello-node...
```

```
| ADDON NAME                  | PROFILE   | STATUS          | MAINTAINER                                |
|---|---|---|---|
| ambassador                  | minikube  | disabled        | 3rd party (Ambassador)                    |
| amd-gpu-device-plugin       | minikube  | disabled        | 3rd party (AMD)                           |
| auto-pause                  | minikube  | disabled        | minikube                                  |
| cloud-spanner               | minikube  | disabled        | Google                                    |
| csi-hostpath-driver         | minikube  | disabled        | Kubernetes                                |
| dashboard                   | minikube  | enabled ✅      | Kubernetes                                |
| default-storageclass        | minikube  | enabled ✅      | Kubernetes                                |
| efk                         | minikube  | disabled        | 3rd party (Elastic)                       |
| freshpod                    | minikube  | disabled        | Google                                    |
| gcp-auth                    | minikube  | disabled        | Google                                    |
| gvisor                      | minikube  | disabled        | minikube                                  |
| headlamp                    | minikube  | disabled        | 3rd party (kinvolk.io)                    |
| inaccel                     | minikube  | disabled        | 3rd party (InAccel [info@inaccel.com])    |
| ingress                     | minikube  | disabled        | Kubernetes                                |
| ingress-dns                 | minikube  | disabled        | minikube                                  |
| inspektor-gadget            | minikube  | disabled        | 3rd party (inspektor-gadget.io)           |
| istio                       | minikube  | disabled        | 3rd party (Istio)                         |
| istio-provisioner           | minikube  | disabled        | 3rd party (Istio)                         |
| kong                        | minikube  | disabled        | 3rd party (Kong HQ)                       |
| kubeflow                    | minikube  | disabled        | 3rd party                                 |
| kubetail                    | minikube  | disabled        | 3rd party (kubetail.com)                  |
| kubevirt                    | minikube  | disabled        | 3rd party (KubeVirt)                      |
| logviewer                   | minikube  | disabled        | 3rd party (unknown)                       |
| metallb                     | minikube  | disabled        | 3rd party (MetalLB)                       |
| metrics-server              | minikube  | disabled        | Kubernetes                                |
| nvidia-device-plugin        | minikube  | disabled        | 3rd party (NVIDIA)                        |
| nvidia-driver-installer     | minikube  | disabled        | 3rd party (NVIDIA)                        |
| nvidia-gpu-device-plugin    | minikube  | disabled        | 3rd party (NVIDIA)                        |
| olm                         | minikube  | disabled        | 3rd party (Operator Framework)            |
| pod-security-policy         | minikube  | disabled        | 3rd party (unknown)                       |
| portainer                   | minikube  | disabled        | 3rd party (Portainer.io)                  |
| registry                    | minikube  | disabled        | minikube                                  |
| registry-aliases            | minikube  | disabled        | 3rd party (unknown)                       |
| registry-creds              | minikube  | disabled        | 3rd party (UPMC Enterprises)              |
| storage-provisioner         | minikube  | enabled ✅      | minikube                                  |
| storage-provisioner-rancher | minikube  | disabled        | 3rd party (Rancher)                       |
| volcano                     | minikube  | disabled        | third-party (volcano)                     |
| volumesnapshots             | minikube  | disabled        | Kubernetes                                |
| yakd                        | minikube  | disabled        | 3rd party (marcnuri.com)                  |
```

```
PS C:\Users\36237> minikube addons enable metrics-server
* metrics-server 是由 Kubernetes 维护的插件。如有任何问题，请在 GitHub 上联系 minikube。
您可以在以下链接查看 minikube 的维护者列表：https://github.com/kubernetes/minikube/blob/master/OWNERS
  - 正在使用镜像 registry.k8s.io/metrics-server/metrics-server:v0.8.1
* 启动 'metrics-server' 插件
PS C:\Users\36237> kubectl get pod,svc -n kube-system
NAME                                       READY   STATUS    RESTARTS   AGE
pod/coredns-7d764666f9-7sffv               1/1     Running   0          34m
pod/etcd-minikube                          1/1     Running   0          34m
pod/kube-apiserver-minikube                1/1     Running   0          34m
pod/kube-controller-manager-minikube       1/1     Running   0          34m
pod/kube-proxy-bgs2s                       1/1     Running   0          34m
pod/kube-scheduler-minikube                1/1     Running   0          34m
pod/metrics-server-9d74bb658-5xsgf         0/1     Running   0          25s
pod/storage-provisioner                    1/1     Running   0          34m

NAME                  TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)                  AGE
service/kube-dns      ClusterIP   10.96.0.10     <none>        53/UDP,53/TCP,9153/TCP   34m
service/metrics-server ClusterIP  10.96.87.33    <none>        443/TCP                  25s
```

```
PS C:\Users\36237> kubectl top pods
error: Metrics API not available
PS C:\Users\36237> kubectl top pods
NAME                          CPU(cores)   MEMORY(bytes)
hello-node-64fc5894d-9zpm8    1m           6Mi
PS C:\Users\36237> minikube addons disable metrics-server
* 'metrics-server' 插件已被禁用
```

3. Deploying an Application

3.1 Understanding Deployments

After verifying the cluster was operational, an application was deployed. In Kubernetes, applications are not run directly as individual containers. Instead, they are managed through higher-level abstractions called Deployments. A Deployment provides declarative updates for pods and replica sets, meaning the user describes the desired end state, and Kubernetes figures out how to achieve and maintain that state.

The Deployment created in this lab used the agnhost test image maintained by the Kubernetes project. This image was instructed to run the netexec command, which starts a simple HTTP server on port 8080. When accessed, this server returns information about the pod including the hostname (which is the pod name), the pod's IP address, and request details.

3.2 Observing the Deployment Process

After creating the deployment, the cluster state was examined. The deployment showed one replica was ready, up-to-date, and available. The single pod created by the deployment was in the Running state, indicating the container started successfully. The pod logs confirmed that the HTTP server had started correctly on port 8080 and the UDP server on port 8081.

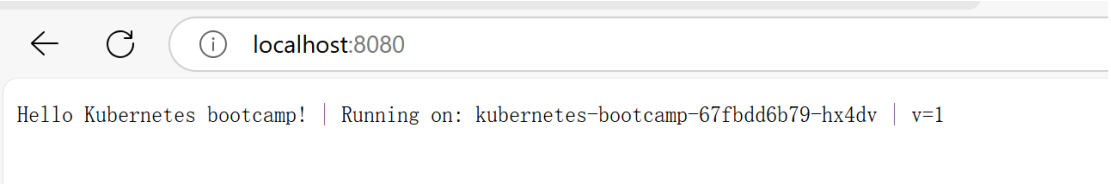
```
PS C:\Users\36237> kubectl create deployment kubernetes-bootcamp --image=gcr.io/google-samples/kubernetes-bootcamp:v1
deployment.apps/kubernetes-bootcamp created
PS C:\Users\36237> kubectl get deployments
NAME                   READY   UP-TO-DATE   AVAILABLE   AGE
kubernetes-bootcamp    0/1     1            0           21s
```

```
StatusCode        : 200
StatusDescription : OK
Content           : {
                      "major": "1",
                      "minor": "35",
                      "emulationMajor": "1",
                      "emulationMinor": "35",
                      "minCompatibilityMajor": "1",
                      "minCompatibilityMinor": "34",
                      "gitVersion": "v1.35.1",
                      "gitCommit": "8fea90...
RawContent        : HTTP/1.1 200 OK
                    Audit-Id: af84ec8a-a665-4ea0-af82-9bf3bbfc9b3c
                    X-Kubernetes-Pf-Flowschema-Uid: eb94c927-f0f6-404d-a282-2812772c6c1e
                    X-Kubernetes-Pf-Prioritylevel-Uid: 662f3168-369a-4205-ab23-4d84a5...
Forms             : {}
Headers           : {[Audit-Id, af84ec8a-a665-4ea0-af82-9bf3bbfc9b3c], [X-Kubernetes-Pf-Flowschema-Uid, eb94c927-f0f6-4
                    04d-a282-2812772c6c1e], [X-Kubernetes-Pf-Prioritylevel-Uid, 662f3168-369a-4205-ab23-4d84a55b7c18],
                    [Content-Length, 379]...}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 379
```

```
PS C:\Users\36237> $POD_NAME = (kubectl get pods -o jsonpath="{.items[0].metadata.name}")
PS C:\Users\36237> echo $POD_NAME
kubernetes-bootcamp-67fbdd6b79-hx4dv
PS C:\Users\36237> curl http://localhost:8001/api/v1/namespaces/default/pods/$POD_NAME:8080/proxy/
curl : { "kind": "Status", "apiVersion": "v1", "metadata": {}, "status": "Failure", "message": "pods \"proxy\" not foun
d", "reason": "NotFound", "details": { "name": "proxy", "kind": "pods" }, "code": 404 }
所在位置 行:1 字符: 1
+ curl http://localhost:8001/api/v1/namespaces/default/pods/$POD_NAME:8 ...
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : InvalidOperation: (System.Net.HttpWebRequest:HttpWebRequest) [Invoke-WebRequest], WebExce
    ption
    + FullyQualifiedErrorId : WebCmdletWebResponseException,Microsoft.PowerShell.Commands.InvokeWebRequestCommand
```

There is a problem， we change another way for it:

```
PS C:\Users\36237> kubectl port-forward $POD_NAME 8080:8080
Forwarding from 127.0.0.1:8080 -> 8080
Forwarding from [::1]:8080 -> 8080
Handling connection for 8080
Handling connection for 8080
```



4. Exposing the Application

4.1 Understanding Services

After deployment, the pod was running but not accessible from outside the cluster. Pod IP addresses are only reachable within the cluster network and are not stable because pods can be terminated and recreated with new IP addresses. To solve these problems, Kubernetes provides an abstraction called a Service. A Service provides a stable, persistent network endpoint for accessing a set of pods. The Service has its own IP address that does not change, and it forwards traffic to the pods that match its selector.

The Service created in this lab was of type LoadBalancer. In cloud environments, this would provision a cloud load balancer with a public IP. In Minikube, the EXTERNAL-IP remains "pending", but the minikube service command provides a workaround by creating a tunnel from the local machine to the Service.

4.2 Accessing the Service

The minikube service command was used to access the application. This command finds the NodePort allocated for the Service, starts a proxy process, and automatically opens a browser window. When the browser made a request, the response showed the pod hostname, pod IP address, and request information, confirming that the Service was correctly routing traffic to the pod.

```
PS C:\Users\36237> kubectl get pods
NAME                                   READY   STATUS    RESTARTS   AGE
kubernetes-bootcamp-67fbdd6b79-hx4dv   1/1     Running   0          24m
PS C:\Users\36237> kubectl describe pods
Name:             kubernetes-bootcamp-67fbdd6b79-hx4dv
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Sun, 19 Apr 2026 19:02:10 +0800
Labels:           app=kubernetes-bootcamp
                  pod-template-hash=67fbdd6b79
Annotations:      <none>
Status:           Running
IP:               10.244.0.10
IPs:
  IP:           10.244.0.10
Controlled By:  ReplicaSet/kubernetes-bootcamp-67fbdd6b79
Containers:
  kubernetes-bootcamp:
    Container ID:   docker://835e13c0416641cd10d70abbc748087f9c43db2b242acf58eb6fdbad55ce39a3
    Image:          gcr.io/google-samples/kubernetes-bootcamp:v1
    Image ID:       docker-pullable://gcr.io/google-samples/kubernetes-bootcamp@sha256:0d6b8ee63bb57c5f5b6156f446b3bc3b3c143d233037f3a2f00e279c8fcc6
4af
    Port:           <none>
    Host Port:      <none>
    State:          Running
      Started:      Sun, 19 Apr 2026 19:02:40 +0800
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-sn7d2 (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-sn7d2:
```

```
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-sn7d2:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s

Events:
  Type    Reason     Age   From               Message
  ----    ------     ----  ----               -------
  Normal  Scheduled  25m   default-scheduler  Successfully assigned default/kubernetes-bootcamp-67fbdd6b79-hx4dv to minikube
  Normal  Pulling    25m   kubelet            Pulling image "gcr.io/google-samples/kubernetes-bootcamp:v1"
  Normal  Pulled     24m   kubelet            Successfully pulled image "gcr.io/google-samples/kubernetes-bootcamp:v1" in 29.058s (29.058s including
 waiting). Image size: 211336459 bytes.
  Normal  Created    24m   kubelet            Container created
  Normal  Started    24m   kubelet            Container started
PS C:\Users\36237>
```

```
PS C:\Users\36237> kubectl exec "$POD_NAME" -- env
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=kubernetes-bootcamp-67fbdd6b79-hx4dv
KUBERNETES_SERVICE_HOST=10.96.0.1
KUBERNETES_SERVICE_PORT=443
KUBERNETES_SERVICE_PORT_HTTPS=443
KUBERNETES_PORT=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP_PROTO=tcp
KUBERNETES_PORT_443_TCP_PORT=443
KUBERNETES_PORT_443_TCP_ADDR=10.96.0.1
NPM_CONFIG_LOGLEVEL=info
NODE_VERSION=6.3.1
HOME=/root
```

```
root@kubernetes-bootcamp-67fbdd6b79-hx4dv:/# cat server.js
var http = require('http');
var requests=0;
var podname= process.env.HOSTNAME;
var startTime;
var host;
var handleRequest = function(request, response) {
  response.setHeader('Content-Type', 'text/plain');
  response.writeHead(200);
  response.write("Hello Kubernetes bootcamp! | Running on: ");
  response.write(host);
  response.end(" | v=1\n");
  console.log("Running On:" ,host, "| Total Requests:", ++requests,"| App Uptime:", (new Date() - startTime)/1000 , "seconds", "| Log Time:",new Dat
e());
}
var www = http.createServer(handleRequest);
www.listen(8080,function () {
    startTime = new Date();;
    host = process.env.HOSTNAME;
    console.log ("Kubernetes Bootcamp App Started At:",startTime, "| Running On: " ,host, "\n" );
});
root@kubernetes-bootcamp-67fbdd6b79-hx4dv:/# curl http://localhost:8080
Hello Kubernetes bootcamp! | Running on: kubernetes-bootcamp-67fbdd6b79-hx4dv | v=1
root@kubernetes-bootcamp-67fbdd6b79-hx4dv:/#
```

5. Exploring the Kubernetes Dashboard

5.1 Launching the Dashboard

The Kubernetes Dashboard is a web-based user interface that provides a visual way to manage and monitor Kubernetes clusters. While kubectl is excellent for scripting and quick operations, the dashboard offers advantages for understanding cluster state by presenting information visually and providing real-time updates.

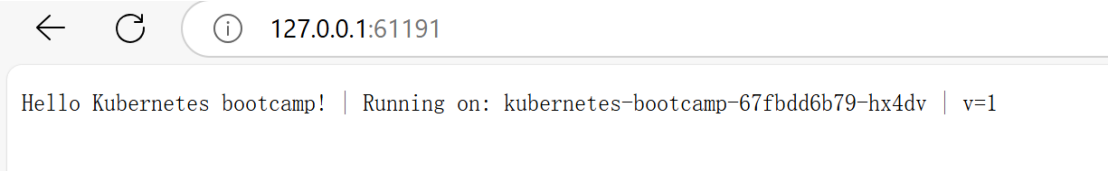
The dashboard was launched and accessed through a browser. The URL output showed a localhost address with a randomly assigned port number, which is a proxy created by minikube to securely forward traffic to the dashboard pod running inside the cluster.

5.2 Dashboard Navigation

Once authenticated, the dashboard presented a comprehensive view of the cluster. The left sidebar contained navigation links to different resource types. The Workloads section revealed the deployment that had been created, showing detailed information including the pod template, the replica set managing the pods, and the current status of each replica.

The dashboard also provided log viewing capabilities, allowing console output from containers to be displayed without using kubectl commands. This visual interface proved particularly useful when observing scaling operations later in the lab.

```
exit
PS C:\Users\36237> kubectl get pods
NAME                                   READY   STATUS    RESTARTS   AGE
kubernetes-bootcamp-67fbdd6b79-hx4dv   1/1     Running   0          30m
PS C:\Users\36237> kubectl get services
NAME         TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
kubernetes   ClusterIP   10.96.0.1    <none>        443/TCP   80m
PS C:\Users\36237> kubectl expose deployment/kubernetes-bootcamp --type="NodePort" --port 8080
service/kubernetes-bootcamp exposed
PS C:\Users\36237> kubectl describe services/kubernetes-bootcamp
Name:                     kubernetes-bootcamp
Namespace:                default
Labels:                   app=kubernetes-bootcamp
Annotations:              <none>
Selector:                 app=kubernetes-bootcamp
Type:                     NodePort
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.98.29.14
IPs:                      10.98.29.14
Port:                     <unset>  8080/TCP
TargetPort:               8080/TCP
NodePort:                 <unset>  31995/TCP
Endpoints:                10.244.0.10:8080
Session Affinity:         None
External Traffic Policy:  Cluster
Internal Traffic Policy:  Cluster
Events:                   <none>
PS C:\Users\36237> export NODE_PORT="$(kubectl get services/kubernetes-bootcamp -o go-template='{{(index .spec.ports 0).nodePort}}')"
export : 无法将“export”项识别为 cmdlet、函数、脚本文件或可运行程序的名称。请检查名称的拼写，如果包括路径，请确保路径正确，然后再试一次。
所在位置 行:1 字符: 1
+ export NODE_PORT="$(kubectl get services/kubernetes-bootcamp -o go-te ...
+ ~~~~~~
    + CategoryInfo          : ObjectNotFound: (export:String) [], CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\36237> $NODE_PORT = kubectl get service/kubernetes-bootcamp -o go-template='{{(index .spec.ports 0).nodePort}}'
PS C:\Users\36237> echo $NODE_PORT
31995
```




```
PS C:\Users\36237> minikube service kubernetes-bootcamp --url
http://127.0.0.1:61191
! 因为你正在使用 windows 上的 Docker 驱动程序，所以需要打开终端才能运行它。
```

```
PS C:\Users\36237> kubectl describe deployment
Name:                   kubernetes-bootcamp
Namespace:              default
CreationTimestamp:      Sun, 19 Apr 2026 19:02:10 +0800
Labels:                 app=kubernetes-bootcamp
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=kubernetes-bootcamp
Replicas:               1 desired | 1 updated | 1 total | 1 available | 0 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=kubernetes-bootcamp
  Containers:
   kubernetes-bootcamp:
    Image:         gcr.io/google-samples/kubernetes-bootcamp:v1
    Port:          <none>
    Host Port:     <none>
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets:  <none>
NewReplicaSet:   kubernetes-bootcamp-67fbdd6b79 (1/1 replicas created)
Events:
  Type    Reason             Age   From                   Message
  ----    ------             ----  ----                   -------
  Normal  ScalingReplicaSet  39m   deployment-controller  Scaled up replica set kubernetes-bootcamp-67fbdd6b79 from 0 to 1
```

```
PS C:\Users\36237> kubectl get pods -l app=kubernetes-bootcamp
NAME                                   READY   STATUS    RESTARTS   AGE
kubernetes-bootcamp-67fbdd6b79-hx4dv   1/1     Running   0          39m
PS C:\Users\36237> kubectl get services -l app=kubernetes-bootcamp
NAME                  TYPE       CLUSTER-IP    EXTERNAL-IP   PORT(S)          AGE
kubernetes-bootcamp   NodePort   10.98.29.14   <none>        8080:31995/TCP   9m41s
PS C:\Users\36237> $NODE_PORT = $(kubectl get pods -o go-template --template '{{range .items}}{{.metadata.name}}{{"\n"}}{{end}}')
error: error parsing template {{range .items}}{{.metadata.name}}{{{\n}}{{end}}, template: output:1: unexpected "\\" in command
PS C:\Users\36237> $POD_NAME = kubectl get pods -o jsonpath="{.items[0].metadata.name}"
PS C:\Users\36237> Write-Host "Name of the Pod: $POD_NAME"
Name of the Pod: kubernetes-bootcamp-67fbdd6b79-hx4dv
```

```
PS C:\Users\36237> kubectl label pods "$POD_NAME" version=v1
pod/kubernetes-bootcamp-67fbdd6b79-hx4dv labeled
PS C:\Users\36237> kubectl describe pods "$POD_NAME"
Name:             kubernetes-bootcamp-67fbdd6b79-hx4dv
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Sun, 19 Apr 2026 19:02:10 +0800
Labels:           app=kubernetes-bootcamp
                  pod-template-hash=67fbdd6b79
                  version=v1
Annotations:      <none>
Status:           Running
IP:               10.244.0.10
IPs:
  IP:           10.244.0.10
Controlled By:  ReplicaSet/kubernetes-bootcamp-67fbdd6b79
Containers:
  kubernetes-bootcamp:
    Container ID:   docker://835e13c0416641cd10d70abbc748087f9c43db2b242acf58eb6fdbad55ce39a3
    Image:          gcr.io/google-samples/kubernetes-bootcamp:v1
    Image ID:       docker-pullable://gcr.io/google-samples/kubernetes-bootcamp@sha256:0d6b8ee63bb57c5f5b6156f446b3bc3b3c143d233037f3a2f00e279c8fcc6
4af
    Port:           <none>
    Host Port:      <none>
    State:          Running
      Started:      Sun, 19 Apr 2026 19:02:40 +0800
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-sn7d2 (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
```

```
Volumes:
  kube-api-access-sn7d2:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s

Events:
  Type    Reason     Age   From               Message
  ----    ------     ----  ----               -------
  Normal  Scheduled  42m   default-scheduler  Successfully assigned default/kubernetes-bootcamp-67fbdd6b79-hx4dv to minikube
  Normal  Pulling    42m   kubelet            Pulling image "gcr.io/google-samples/kubernetes-bootcamp:v1"
  Normal  Pulled     42m   kubelet            Successfully pulled image "gcr.io/google-samples/kubernetes-bootcamp:v1" in 29.058s (29.058s including
 waiting). Image size: 211336459 bytes.
  Normal  Created    42m   kubelet            Container created
  Normal  Started    42m   kubelet            Container started
PS C:\Users\36237>
```

```
PS C:\Users\36237> kubectl get pods -l version=v1
NAME                                   READY   STATUS    RESTARTS   AGE
kubernetes-bootcamp-67fbdd6b79-hx4dv   1/1     Running   0          43m
PS C:\Users\36237>
```

```
PS C:\Users\36237> kubectl delete service -l app=kubernetes-bootcamp
service "kubernetes-bootcamp" deleted from default namespace
PS C:\Users\36237> kubectl get services
NAME         TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
kubernetes   ClusterIP   10.96.0.1    <none>        443/TCP   94m
```

6. Scaling the Application

6.1 Horizontal Scaling

One of Kubernetes' most powerful features is horizontal scaling, which means increasing the number of running instances of an application. This is generally preferred over vertical scaling (increasing resources for a single instance) because it allows applications to handle increased load by adding more instances rather than needing larger servers.

The deployment was scaled from one replica to three replicas. When the scale command was issued, the Deployment controller noticed the desired state change from 1 to 3 replicas. The controller updated the ReplicaSet to request 3 replicas, and the ReplicaSet controller created two new pod objects. The scheduler assigned these pods to the available node, and the kubelet started the new containers. The entire scaling process took only a few seconds.

6.2 Verifying the Scale Operation

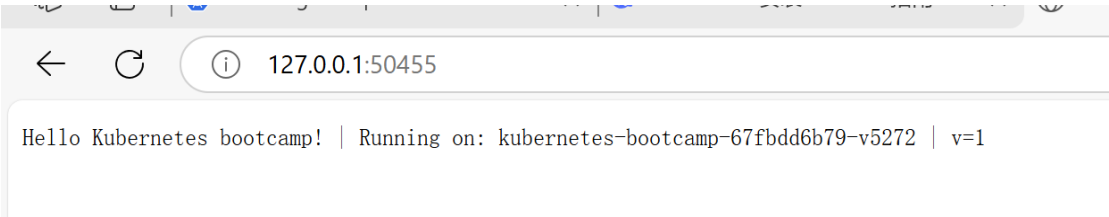
After scaling, three pods were shown instead of one. All three pods were in the Running state. The Service automatically started load balancing across all three pods without any additional configuration. When multiple pods exist, the Service distributes traffic among them using a round-robin algorithm. This automatic load balancing is a key benefit of using Kubernetes Services.

```
PS C:\Users\36237> kubectl expose deployment/kubernetes-bootcamp --type="LoadBalancer" --port 8080
service/kubernetes-bootcamp exposed
PS C:\Users\36237> kubectl get deployments
NAME                   READY   UP-TO-DATE   AVAILABLE   AGE
kubernetes-bootcamp    1/1     1            1           50m
PS C:\Users\36237> kubectl get rs
NAME                             DESIRED   CURRENT   READY   AGE
kubernetes-bootcamp-67fbdd6b79   1         1         1       51m
PS C:\Users\36237> kubectl scale deployments/kubernetes-bootcamp --replicas=4
deployment.apps/kubernetes-bootcamp scaled
PS C:\Users\36237> kubectl get deployments
NAME                   READY   UP-TO-DATE   AVAILABLE   AGE
kubernetes-bootcamp    4/4     4            4           51m
PS C:\Users\36237> kubectl get pods -o wide
NAME                                   READY   STATUS    RESTARTS   AGE   IP            NODE       NOMINATED NODE   READINESS GATES
kubernetes-bootcamp-67fbdd6b79-hx4dv   1/1     Running   0          51m   10.244.0.10   minikube   <none>           <none>
kubernetes-bootcamp-67fbdd6b79-r9rp7   1/1     Running   0          23s   10.244.0.13   minikube   <none>           <none>
kubernetes-bootcamp-67fbdd6b79-t48qq   1/1     Running   0          23s   10.244.0.12   minikube   <none>           <none>
kubernetes-bootcamp-67fbdd6b79-v5272   1/1     Running   0          23s   10.244.0.11   minikube   <none>           <none>
PS C:\Users\36237> kubectl describe deployments/kubernetes-bootcamp
Name:                   kubernetes-bootcamp
Namespace:              default
CreationTimestamp:      Sun, 19 Apr 2026 19:02:10 +0800
Labels:                 app=kubernetes-bootcamp
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=kubernetes-bootcamp
Replicas:               4 desired | 4 updated | 4 total | 4 available | 0 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=kubernetes-bootcamp
  Containers:
   kubernetes-bootcamp:
    Image:         gcr.io/google-samples/kubernetes-bootcamp:v1
    Port:          <none>
    Host Port:     <none>
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
```

```
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Progressing    True    NewReplicaSetAvailable
  Available      True    MinimumReplicasAvailable
OldReplicaSets:  <none>
NewReplicaSet:   kubernetes-bootcamp-67fbdd6b79 (4/4 replicas created)
Events:
  Type    Reason             Age   From                   Message
  ----    ------             ----  ----                   -------
  Normal  ScalingReplicaSet  51m   deployment-controller  Scaled up replica set kubernetes-bootcamp-67fbdd6b79 from 0 to 1
  Normal  ScalingReplicaSet  33s   deployment-controller  Scaled up replica set kubernetes-bootcamp-67fbdd6b79 from 1 to 4
PS C:\Users\36237>
```

```
PS C:\Users\36237> kubectl describe services/kubernetes-bootcamp
Name:                     kubernetes-bootcamp
Namespace:                default
Labels:                   app=kubernetes-bootcamp
Annotations:              <none>
Selector:                 app=kubernetes-bootcamp
Type:                     LoadBalancer
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.107.251.2
IPs:                      10.107.251.2
Port:                     <unset>  8080/TCP
TargetPort:               8080/TCP
NodePort:                 <unset>  30098/TCP
Endpoints:                10.244.0.10:8080,10.244.0.12:8080,10.244.0.11:8080 + 1 more...
Session Affinity:         None
External Traffic Policy:  Cluster
Internal Traffic Policy:  Cluster
Events:                   <none>
```

```
PS C:\Users\36237> kubectl port-forward service/kubernetes-bootcamp 8080:8080
Forwarding from 127.0.0.1:8080 -> 8080
Forwarding from [::1]:8080 -> 8080
Handling connection for 8080
Handling connection for 8080
PS C:\Users\36237> minikube service kubernetes-bootcamp --url
http://127.0.0.1:50455
! 因为你正在使用 windows 上的 Docker 驱动程序，所以需要打开终端才能运行它。
```




```
PS C:\Users\36237> kubectl scale deployments/kubernetes-bootcamp --replicas=2
deployment.apps/kubernetes-bootcamp scaled
PS C:\Users\36237> kubectl get deployments
NAME                  READY   UP-TO-DATE   AVAILABLE   AGE
kubernetes-bootcamp   2/2     2            2           66m
PS C:\Users\36237> kubectl get pods -o wide
NAME                                   READY   STATUS        RESTARTS   AGE   IP            NODE       NOMINATED NODE   READINESS GATES
kubernetes-bootcamp-67fbdd6b79-hx4dv   1/1     Running       0          66m   10.244.0.10   minikube   <none>           <none>
kubernetes-bootcamp-67fbdd6b79-r9rp7   1/1     Terminating   0          15m   10.244.0.13   minikube   <none>           <none>
kubernetes-bootcamp-67fbdd6b79-t48qq   1/1     Running       0          15m   10.244.0.12   minikube   <none>           <none>
kubernetes-bootcamp-67fbdd6b79-v5272   1/1     Terminating   0          15m   10.244.0.11   minikube   <none>           <none>
PS C:\Users\36237>
```

7. Updating the Application

7.1 Rolling Updates

The final major task was to update the application to a new version. Kubernetes solves the problem of update downtime through rolling updates. A rolling update gradually replaces old pods with new ones. Throughout the update process, the Service continues to send traffic only to pods that are healthy and

ready, resulting in zero-downtime updates.

When the update command was issued to change the image tag from 2.39 to 2.40, the Deployment controller created a new ReplicaSet with the new pod template. It then performed the rolling update by creating one new pod while keeping all three old pods running. Once the new pod passed its readiness checks, one old pod was terminated. This process repeated until all old pods were replaced by new pods.

7.2 Observing the Update

The rollout status command showed the progress of the update, with messages indicating the deployment was waiting for old replicas to be terminated and then confirming successful completion. The rollout history command showed two revisions of the deployment: revision 1 with the original image and revision 2 with the updated image.

After the update completed, the pod names had changed. The hash in the pod names was different, indicating these were new pods created from the updated pod template. The old pods had been terminated and removed from the cluster. All new pods showed a Running status, confirming the new container image started successfully.

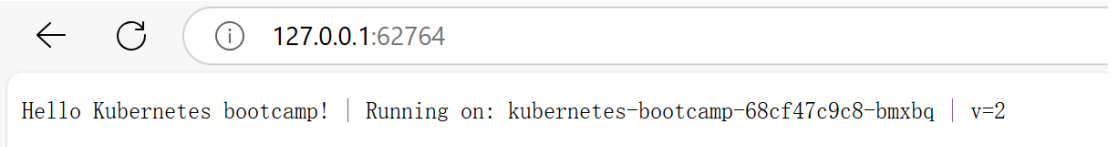
```
PS C:\Users\36237> kubectl get deployments
NAME                   READY   UP-TO-DATE   AVAILABLE   AGE
kubernetes-bootcamp    2/2     2            2           68m
PS C:\Users\36237> kubectl get pods
NAME                                   READY   STATUS    RESTARTS   AGE
kubernetes-bootcamp-67fbdd6b79-hx4dv   1/1     Running   0          68m
kubernetes-bootcamp-67fbdd6b79-t48qq   1/1     Running   0          17m
PS C:\Users\36237> kubectl describe pods
Name:             kubernetes-bootcamp-67fbdd6b79-hx4dv
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Sun, 19 Apr 2026 19:02:10 +0800
Labels:           app=kubernetes-bootcamp
                  pod-template-hash=67fbdd6b79
                  version=v1
Annotations:      <none>
Status:           Running
IP:               10.244.0.10
IPs:
  IP:           10.244.0.10
Controlled By:  ReplicaSet/kubernetes-bootcamp-67fbdd6b79
Containers:
  kubernetes-bootcamp:
    Container ID:   docker://835e13c0416641cd10d70abbc748087f9c43db2b242acf58eb6fdbad55ce39a3
    Image:          gcr.io/google-samples/kubernetes-bootcamp:v1
    Image ID:       docker-pullable://gcr.io/google-samples/kubernetes-bootcamp@sha256:0d6b8ee63bb57c5f5b6156f446b3bc3b3c143d233037f3a2f00e279c8fcc6
4af
    Port:           <none>
    Host Port:      <none>
    State:          Running
      Started:      Sun, 19 Apr 2026 19:02:40 +0800
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-sn7d2 (ro)
Conditions:
```

```
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-sn7d2:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:                      <none>


Name:             kubernetes-bootcamp-67fbdd6b79-t48qq
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Sun, 19 Apr 2026 19:53:31 +0800
Labels:           app=kubernetes-bootcamp
                  pod-template-hash=67fbdd6b79
Annotations:      <none>
Status:           Running
IP:               10.244.0.12
IPs:
  IP:           10.244.0.12
Controlled By:  ReplicaSet/kubernetes-bootcamp-67fbdd6b79
Containers:
  kubernetes-bootcamp:
    Container ID:   docker://cec19798cffcff135bdf4f1c28bce60ac9f6eddf59437ac32ad2752113dd6715
    Image:          gcr.io/google-samples/kubernetes-bootcamp:v1
    Image ID:       docker-pullable://gcr.io/google-samples/kubernetes-bootcamp@sha256:0d6b8ee63bb57c5f5b6156f446b3bc3b3c143d233037f3a2f00e279c8fcc6
4af
```

```
4af
    Port:           <none>
    Host Port:      <none>
    State:          Running
      Started:      Sun, 19 Apr 2026 19:53:32 +0800
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-ph4zp (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-ph4zp:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s

Events:
  Type    Reason     Age   From               Message
  ----    ------     ----  ----               -------
  Normal  Scheduled  17m   default-scheduler  Successfully assigned default/kubernetes-bootcamp-67fbdd6b79-t48qq to minikube
  Normal  Pulled     17m   kubelet            Container image "gcr.io/google-samples/kubernetes-bootcamp:v1" already present on machine and can be a
ccessed by the pod
  Normal  Created    17m   kubelet            Container created
  Normal  Started    17m   kubelet            Container started
```

```
PS C:\Users\36237> kubectl set image deployments/kubernetes-bootcamp kubernetes-bootcamp=docker.io/jocatalin/kubernetes-bootcamp:v2
deployment.apps/kubernetes-bootcamp image updated
PS C:\Users\36237> kubectl get pods
NAME                                   READY   STATUS              RESTARTS   AGE
kubernetes-bootcamp-67fbdd6b79-hx4dv   1/1     Running             0          69m
kubernetes-bootcamp-67fbdd6b79-t48qq   1/1     Running             0          18m
kubernetes-bootcamp-68cf47c9c8-mn6c5   0/1     ContainerCreating   0          11s
```

```
PS C:\Users\36237> kubectl set image deployments/kubernetes-bootcamp kubernetes-bootcamp=docker.io/jocatalin/kubernetes-bootcamp:v2
deployment.apps/kubernetes-bootcamp image updated
PS C:\Users\36237> kubectl get pods
NAME                                   READY   STATUS              RESTARTS   AGE
kubernetes-bootcamp-67fbdd6b79-hx4dv   1/1     Running             0          69m
kubernetes-bootcamp-67fbdd6b79-t48qq   1/1     Running             0          18m
kubernetes-bootcamp-68cf47c9c8-mn6c5   0/1     ContainerCreating   0          11s
PS C:\Users\36237> kubectl expose deployment/kubernetes-bootcamp --type="NodePort" --port 8080
Error from server (AlreadyExists): services "kubernetes-bootcamp" already exists
PS C:\Users\36237> $NODE_PORT = kubectl get services/kubernetes-bootcamp -o go-template='{{(index .spec.ports 0).nodePort}}'
PS C:\Users\36237> Write-Host "NODE_PORT=$NODE_PORT"
NODE_PORT=30098
PS C:\Users\36237>  minikube service kubernetes-bootcamp --url
http://127.0.0.1:62764
! 因为你正在使用 windows 上的 Docker 驱动程序，所以需要打开终端才能运行它。
```




```
! 因为你正在使用 windows 上的 Docker 驱动程序，所以需要打开终端才能运行它。
PS C:\Users\36237> kubectl rollout status deployments/kubernetes-bootcamp
deployment "kubernetes-bootcamp" successfully rolled out
PS C:\Users\36237> kubectl describe pods
Name:             kubernetes-bootcamp-68cf47c9c8-bmxbq
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Sun, 19 Apr 2026 20:11:54 +0800
Labels:           app=kubernetes-bootcamp
                  pod-template-hash=68cf47c9c8
Annotations:      <none>
Status:           Running
IP:               10.244.0.15
IPs:
  IP:           10.244.0.15
Controlled By:  ReplicaSet/kubernetes-bootcamp-68cf47c9c8
Containers:
  kubernetes-bootcamp:
    Container ID:   docker://fbdf92a81ad17a1f816347e3ab1012de9fd9fd556e3cbcd435b83081ce3be9fc
    Image:          docker.io/jocatalin/kubernetes-bootcamp:v2
    Image ID:       docker-pullable://jocatalin/kubernetes-bootcamp@sha256:fb1a3ced00cecfc1f83f18ab5cd14199e30adc1b49aa4244f5d65ad3f5feb2a5
    Port:           <none>
    Host Port:      <none>
    State:          Running
      Started:      Sun, 19 Apr 2026 20:11:55 +0800
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-fn2rv (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-fn2rv:
```

```
  kube-api-access-fn2rv:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason     Age    From               Message
  ----    ------     ----   ----               -------
  Normal  Scheduled  3m20s  default-scheduler  Successfully assigned default/kubernetes-bootcamp-68cf47c9c8-bmxbq to minikube
  Normal  Pulled     3m19s  kubelet            Container image "docker.io/jocatalin/kubernetes-bootcamp:v2" already present on machine and can be ac
cessed by the pod
  Normal  Created    3m19s  kubelet            Container created
  Normal  Started    3m19s  kubelet            Container started


Name:             kubernetes-bootcamp-68cf47c9c8-mn6c5
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Sun, 19 Apr 2026 20:11:43 +0800
Labels:           app=kubernetes-bootcamp
                  pod-template-hash=68cf47c9c8
Annotations:      <none>
Status:           Running
IP:               10.244.0.14
IPs:
  IP:           10.244.0.14
Controlled By:  ReplicaSet/kubernetes-bootcamp-68cf47c9c8
Containers:
  kubernetes-bootcamp:
    Container ID:   docker://816d200320887098d615f5c9fe90578037354 3aa325e0cb0c844aae1a490f898
    Image:          docker.io/jocatalin/kubernetes-bootcamp:v2
    Image ID:       docker-pullable://jocatalin/kubernetes-bootcamp@sha256:fb1a3ced00cecfc1f83f18ab5cd14199e30adc1b49aa4244f5d65ad3f5feb2a5
    Port:           <none>
    Host Port:      <none>
    State:          Running
      Started:      Sun, 19 Apr 2026 20:11:54 +0800
```

```
Containers:
  kubernetes-bootcamp:
    Container ID:   docker://816d200320887098d615f5c9fe90578037354 3aa325e0cb0c844aae1a490f898
    Image:          docker.io/jocatalin/kubernetes-bootcamp:v2
    Image ID:       docker-pullable://jocatalin/kubernetes-bootcamp@sha256:fb1a3ced00cecfc1f83f18ab5cd14199e30adc1b49aa4244f5d65ad3f5feb2a5
    Port:           <none>
    Host Port:      <none>
    State:          Running
      Started:      Sun, 19 Apr 2026 20:11:54 +0800
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-r5572 (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-r5572:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason     Age    From               Message
  ----    ------     ----   ----               -------
  Normal  Scheduled  3m31s  default-scheduler  Successfully assigned default/kubernetes-bootcamp-68cf47c9c8-mn6c5 to minikube
  Normal  Pulling    3m30s  kubelet            Pulling image "docker.io/jocatalin/kubernetes-bootcamp:v2"
  Normal  Pulled     3m20s  kubelet            Successfully pulled image "docker.io/jocatalin/kubernetes-bootcamp:v2" in 9.494s (9.494s including wa
iting). Image size: 211336459 bytes.
  Normal  Created    3m20s  kubelet            Container created
  Normal  Started    3m20s  kubelet            Container started
PS C:\Users\36237>
```

```
PS C:\Users\36237> kubectl set image deployments/kubernetes-bootcamp kubernetes-bootcamp=gcr.io/google-samples/kubernetes-bootcamp:v10
deployment.apps/kubernetes-bootcamp image updated
PS C:\Users\36237> kubectl get deployments
NAME                  READY   UP-TO-DATE   AVAILABLE   AGE
kubernetes-bootcamp   2/2     1            2           74m
PS C:\Users\36237> kubectl get pods
NAME                                   READY   STATUS         RESTARTS   AGE
kubernetes-bootcamp-68cf47c9c8-bmxbq   1/1     Running        0          5m5s
kubernetes-bootcamp-68cf47c9c8-mn6c5   1/1     Running        0          5m16s
kubernetes-bootcamp-845bc89d75-ff5rp   0/1     ErrImagePull   0          23s
PS C:\Users\36237> kubectl describe pods
Name:             kubernetes-bootcamp-68cf47c9c8-bmxbq
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Sun, 19 Apr 2026 20:11:54 +0800
Labels:           app=kubernetes-bootcamp
                  pod-template-hash=68cf47c9c8
Annotations:      <none>
Status:           Running
IP:               10.244.0.15
IPs:
  IP:           10.244.0.15
Controlled By:  ReplicaSet/kubernetes-bootcamp-68cf47c9c8
Containers:
  kubernetes-bootcamp:
    Container ID:   docker://fbdf92a81ad17a1f816347e3ab1012de9fd9fd556e3cbcd435b83081ce3be9fc
    Image:          docker.io/jocatalin/kubernetes-bootcamp:v2
    Image ID:       docker-pullable://jocatalin/kubernetes-bootcamp@sha256:fb1a3ced00cecfc1f83f18ab5cd14199e30adc1b49aa4244f5d65ad3f5feb2a5
    Port:           <none>
    Host Port:      <none>
    State:          Running
      Started:      Sun, 19 Apr 2026 20:11:55 +0800
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-fn2rv (ro)
Conditions:
```

```
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-fn2rv:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason     Age    From               Message
  ----    ------     ----   ----               -------
  Normal  Scheduled  5m17s  default-scheduler  Successfully assigned default/kubernetes-bootcamp-68cf47c9c8-bmxbq to minikube
  Normal  Pulled     5m16s  kubelet            Container image "docker.io/jocatalin/kubernetes-bootcamp:v2" already present on machine and can be ac
cessed by the pod
  Normal  Created    5m16s  kubelet            Container created
  Normal  Started    5m16s  kubelet            Container started


Name:             kubernetes-bootcamp-68cf47c9c8-mn6c5
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Sun, 19 Apr 2026 20:11:43 +0800
Labels:           app=kubernetes-bootcamp
                  pod-template-hash=68cf47c9c8
Annotations:      <none>
Status:           Running
IP:               10.244.0.14
IPs:
  IP:           10.244.0.14
Controlled By:  ReplicaSet/kubernetes-bootcamp-68cf47c9c8
Containers:
```

```
Containers:
  kubernetes-bootcamp:
    Container ID:   docker://816d200320887098d615f5c9fe90578037354 3aa325e0cb0c844aae1a490f898
    Image:          docker.io/jocatalin/kubernetes-bootcamp:v2
    Image ID:       docker-pullable://jocatalin/kubernetes-bootcamp@sha256:fb1a3ced00cecfc1f83f18ab5cd14199e30adc1b49aa4244f5d65ad3f5feb2a5
    Port:           <none>
    Host Port:      <none>
    State:          Running
      Started:      Sun, 19 Apr 2026 20:11:54 +0800
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-r5572 (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-r5572:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason     Age    From               Message
  ----    ------     ----   ----               -------
  Normal  Scheduled  5m28s  default-scheduler  Successfully assigned default/kubernetes-bootcamp-68cf47c9c8-mn6c5 to minikube
  Normal  Pulling    5m27s  kubelet            Pulling image "docker.io/jocatalin/kubernetes-bootcamp:v2"
  Normal  Pulled     5m17s  kubelet            Successfully pulled image "docker.io/jocatalin/kubernetes-bootcamp:v2" in 9.494s (9.494s including wa
iting). Image size: 211336459 bytes.
  Normal  Created    5m17s  kubelet            Container created
  Normal  Started    5m17s  kubelet            Container started
```

```
Name:             kubernetes-bootcamp-845bc89d75-ff5rp
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Sun, 19 Apr 2026 20:16:36 +0800
Labels:           app=kubernetes-bootcamp
                  pod-template-hash=845bc89d75
Annotations:      <none>
Status:           Pending
IP:               10.244.0.16
IPs:
  IP:           10.244.0.16
Controlled By:  ReplicaSet/kubernetes-bootcamp-845bc89d75
Containers:
  kubernetes-bootcamp:
    Container ID:
    Image:          gcr.io/google-samples/kubernetes-bootcamp:v10
    Image ID:
    Port:           <none>
    Host Port:      <none>
    State:          Waiting
      Reason:       ImagePullBackOff
    Ready:          False
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-dwndg (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       False
  ContainersReady             False
  PodScheduled                True
Volumes:
  kube-api-access-dwndg:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
    DownwardAPI:             true
```

```
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       False
  ContainersReady             False
  PodScheduled                True
Volumes:
  kube-api-access-dwndg:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s

Events:
  Type     Reason     Age                From               Message
  ----     ------     ----               ----               -------
  Normal   Scheduled  35s                default-scheduler  Successfully assigned default/kubernetes-bootcamp-845bc89d75-ff5rp to minikube
  Warning  Failed     26s                kubelet            Failed to pull image "gcr.io/google-samples/kubernetes-bootcamp:v10": Error response fro
m daemon: manifest for gcr.io/google-samples/kubernetes-bootcamp:v10 not found: manifest unknown: Failed to fetch "v10"
  Warning  Failed     26s                kubelet            Error: ErrImagePull
  Normal   BackOff    25s                kubelet            Back-off pulling image "gcr.io/google-samples/kubernetes-bootcamp:v10"
  Warning  Failed     25s                kubelet            Error: ImagePullBackOff
  Normal   Pulling    11s (x2 over 34s)  kubelet            Pulling image "gcr.io/google-samples/kubernetes-bootcamp:v10"
PS C:\Users\36237>
```

```
PS C:\Users\36237> kubectl rollout undo deployments/kubernetes-bootcamp
deployment.apps/kubernetes-bootcamp rolled back
PS C:\Users\36237> kubectl get pods
NAME                                   READY   STATUS    RESTARTS   AGE
kubernetes-bootcamp-68cf47c9c8-bmxbq   1/1     Running   0          6m38s
kubernetes-bootcamp-68cf47c9c8-mn6c5   1/1     Running   0          6m49s
PS C:\Users\36237> kubectl describe pods
Name:             kubernetes-bootcamp-68cf47c9c8-bmxbq
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Sun, 19 Apr 2026 20:11:54 +0800
Labels:           app=kubernetes-bootcamp
                  pod-template-hash=68cf47c9c8
Annotations:      <none>
Status:           Running
IP:               10.244.0.15
IPs:
  IP:           10.244.0.15
Controlled By:  ReplicaSet/kubernetes-bootcamp-68cf47c9c8
Containers:
  kubernetes-bootcamp:
    Container ID:   docker://fbdf92a81ad17a1f816347e3ab1012de9fd9fd556e3cbcd435b83081ce3be9fc
    Image:          docker.io/jocatalin/kubernetes-bootcamp:v2
    Image ID:       docker-pullable://jocatalin/kubernetes-bootcamp@sha256:fb1a3ced00cecfc1f83f18ab5cd14199e30adc1b49aa4244f5d65ad3f5feb2a5
    Port:           <none>
    Host Port:      <none>
    State:          Running
      Started:      Sun, 19 Apr 2026 20:11:55 +0800
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-fn2rv (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
```

```
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-fn2rv (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-fn2rv:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s

Events:
  Type    Reason     Age    From               Message
  ----    ------     ----   ----               -------
  Normal  Scheduled  6m53s  default-scheduler  Successfully assigned default/kubernetes-bootcamp-68cf47c9c8-bmxbq to minikube
  Normal  Pulled     6m52s  kubelet            Container image "docker.io/jocatalin/kubernetes-bootcamp:v2" already present on machine and can be ac
cessed by the pod
  Normal  Created    6m52s  kubelet            Container created
  Normal  Started    6m52s  kubelet            Container started


Name:             kubernetes-bootcamp-68cf47c9c8-mn6c5
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Sun, 19 Apr 2026 20:11:43 +0800
Labels:           app=kubernetes-bootcamp
                  pod-template-hash=68cf47c9c8
Annotations:      <none>
Status:           Running
IP:               10.244.0.14
```

```
IP:               10.244.0.14
IPs:
  IP:           10.244.0.14
Controlled By:  ReplicaSet/kubernetes-bootcamp-68cf47c9c8
Containers:
  kubernetes-bootcamp:
    Container ID:   docker://816d200320887098d615f5c9fe90578037354 3aa325e0cb0c844aae1a490f898
    Image:          docker.io/jocatalin/kubernetes-bootcamp:v2
    Image ID:       docker-pullable://jocatalin/kubernetes-bootcamp@sha256:fb1a3ced00cecfc1f83f18ab5cd14199e30adc1b49aa4244f5d65ad3f5feb2a5
    Port:           <none>
    Host Port:      <none>
    State:          Running
      Started:      Sun, 19 Apr 2026 20:11:54 +0800
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-r5572 (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-r5572:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason     Age    From               Message
  ----    ------     ----   ----               -------
  Normal  Scheduled  7m4s   default-scheduler  Successfully assigned default/kubernetes-bootcamp-68cf47c9c8-mn6c5 to minikube
  Normal  Pulling    7m3s   kubelet            Pulling image "docker.io/jocatalin/kubernetes-bootcamp:v2"
  Normal  Pulled     6m53s  kubelet            Successfully pulled image "docker.io/jocatalin/kubernetes-bootcamp:v2" in 9.494s (9.494s including wa
iting). Image size: 211336459 bytes.
```

```
PS C:\Users\36237> kubectl delete deployments/kubernetes-bootcamp services/kubernetes-bootcamp
deployment.apps "kubernetes-bootcamp" deleted from default namespace
service "kubernetes-bootcamp" deleted from default namespace
PS C:\Users\36237>
```

8. Enabling Observability with Metrics Server

8.1 Metrics Server Overview

After completing the core tutorial objectives, the metrics-server addon was enabled to add observability capabilities. The metrics-server collects resource metrics from the kubelets on each node and aggregates them. It provides current CPU and memory usage data for nodes and pods.

8.2 Using Metrics

The metrics-server addon was enabled. It took some time for the metrics-server pod to start and begin collecting metrics. Once available, the kubectl top pods command showed CPU and memory usage for each pod. CPU usage was measured in millicores (1000 millicores = 1 CPU core), and memory usage was measured in mebibytes. Each pod was using approximately 1 millicore of CPU and 8 mebibytes of memory, which was expected given the lightweight nature of the agnhost application.

9. Resource Cleanup

9.1 Deleting Resources

After completing all lab objectives, the resources were properly cleaned up. The Service and Deployment were deleted, which automatically removed the associated endpoints, iptables rules, ReplicaSet, and pods. The containers running in those pods were stopped, though the container images remained cached on the node.

9.2 Stopping and Deleting Minikube

The Minikube cluster was stopped, which gracefully shuts down the cluster without deleting it. This is useful when work is temporarily paused. Finally, the cluster was completely removed using minikube delete, which deleted the Docker container, removed cluster configuration from the kubeconfig file, and freed disk space.

10. Key Takeaways

This lab provided hands-on experience with several fundamental Kubernetes concepts:

Declarative Configuration: Instead of executing sequences of commands describing how to achieve a desired state, Kubernetes users describe what the desired state should be, and Kubernetes figures out how to achieve it. This approach is more reliable because it eliminates the need to handle intermediate states manually.

Control Loop Pattern: Controllers continuously watch the actual state of the cluster and take action when it does not match the desired state. This enables self-healing capabilities where Kubernetes automatically responds to failures.

Abstraction Layers: Pods abstract away individual containers, deployments abstract away pod management, and services abstract away networking. Each layer builds on the layers below, providing higher-level functionality while hiding complexity.

11. Conclusion

This lab successfully demonstrated the core concepts of Kubernetes using Minikube. All major operations were completed successfully: cluster setup, application deployment, service exposure, dashboard exploration, scaling, rolling updates, and observability. The skills acquired in this lab provide a solid foundation for more advanced Kubernetes topics including service discovery, persistent storage, ingress controllers, and production cluster management.